

ABRA v2 — The Redesign & the ALAKAZAM Plan

A Champions VGC model family, rebuilt honestly around what the data and the literature actually support. 2026-07-23

1 · The reframe (why we redesigned)

We stress-tested the v1 models against real games and got two decisive results. **First, the good:** the damage engine is **exact** — it matches the trusted @smogon/calc to the integer (within 5% on 100% of tests). **Second, the hard truth:** predicting *who wins a game from team sheets* is near-impossible in this format — even player-Elo, the gold standard, barely beats a coin (log-loss 0.687 vs 0.693). JOLTEON ties a coin; MEDICHAM's win% is actually *below* chance because its greedy policy is systematically biased (it backs fast/offensive teams, which lose more).

The lesson isn't 'the models are bad' — it's that we aimed at an unwinnable target. In a high-variance, non-transitive game, the sheet doesn't decide the match; decisions and variance do. So v2 stops predicting outcomes and starts **supporting decisions** — which is exactly where skill lives, and where value is real.

2 · What the literature says (studied hard)

The Pokémon field already answered this (2024–25). Metamon (UT-Austin, RLC 2025) reaches the **top 10% of human players** using **offline RL on human replays + self-play and no search at all** — beating an LLM agent and a heuristic search engine. **VGC-Bench** (June 2025) confirms doubles VGC is $\sim 10^{139}$ configurations and that the credible strategic method is **PSRO / double-oracle with a Nash mixture**.

Other games solved our sub-problems. **Stratego/DeepNash:** expert play on *hidden identities* (our hidden sets), model-free, via Regularized Nash Dynamics. **StarCraft/AlphaStar:** a league + Nash mixture for robustness, and 'scouting' — *information-gathering has value*. **Diplomacy/CICERO:** the key trick — **maximize expected value but stay KL-anchored to human play**; pure greedy RL drifts into exploitable corners (exactly how MEDICHAM inverted). **Sports analytics (xG/EPV):** an entire field that abandoned predicting noisy outcomes for **expected value of decisions** — this is our thesis, validated.

The convergent recipe (poker + Stratego + StarCraft + Diplomacy all agree): *learn a policy/value by self-play from human data → keep it near human play (KL-anchor) → select strategies by a Nash mixture for robustness → add depth-limited search only to sharpen tactics* — and present everything as calibrated expected value, never as an outcome oracle.

3 · The v2 model family (pivot, don't retire)

Almost nothing is thrown away — the pieces become **components of a decision stack**. New models get names in the ABRA style; only the false *framing* (win-prediction) is retired.

Model	Pokémon — why	Role in v2	Status
MEDICHAM	fighting/psychic — the exact striker	Validated damage engine + rollout simulator + search leaf	FOUNDATION ✓
XATU	foresees future & past	Belief model — opponent hidden sets + next move (I1)	BUILD
PORYGON-Z	man-made code that learns	Value/policy net — the learned brain; line PORYGON→2→Z = imitation→offline-RL→self-play (I2)	BUILD
SLOWKING	the sage	Team-preview Nash — bring/lead mixed strategy (I3)	BUILD
CHOMP	bites down on the right 4	Bring/lead as calibrated EV; prove it beats human brings (I4)	PIVOT + PROVE
GURU	the sage who has watched every battle	Meta/matchup model — matchup matrix from REAL outcomes (I5)	BUILD
HYPNO	reads minds, exploits	Opponent-type / exploitability — sets the piKL anchor (I6)	BUILD
DUSKNOIR	finality	Endgame exact solver — small positions solved to certainty (I7)	BUILD
SMEARGLE	sketches endless games	Self-play data engine + curation — fuel for the net (I8)	BUILD
ROTOM	forme-swaps per appliance	Gimmick module (swappable per reg; currently Mega)	BUILD
JOLTEON	fast, jumpy	Demoted to a fast usage-prior / shortlister (no win claim)	REFRAME
KADABRA → ALAKAZAM	the spoon evolves	Replay coach → the capstone decision engine	CAPSTONE

4 · Dependency graph & build order (inputs first, capstone last)

ALAKAZAM is the last brick — it *consumes* the others, so we secure each input (with an honest metric + confidence interval + baseline, gated in CI) before assembling the capstone.

MEDICHAM (✓) + SMEARGLE data → **XATU** (belief), **PORYGON-Z** (net), **GURU** (matchups), **HYPNO** (opponent), **DUSKNOIR** (endgame) → **SLOWKING** (preview) + **CHOMP** (bring/lead) → **ALAKAZAM** (in-battle capstone).

Build order: (1) MEDICHAM damage ✓ (2) XATU belief + CHOMP-EV proof + ORANGURU matchup (3) PORYGON-Z net (the heavy lift) (4) SLOWKING preview-Nash (5) DITTO→PSRO team-builder (6)

ALAKAZAM.

5 · ALAKAZAM — the capstone (honest definition)

Given a **live position**, ALAKAZAM outputs the **win-%-optimal move and its value** by: (1) forming a **belief** (XATU) over the opponent's hidden info; (2) proposing a move from the **learned policy** (PORYGON-Z), KL-anchored to human play; (3) running **depth-limited search** over the validated engine, solving each simultaneous turn as a **matrix game** (kills the greedy speed-bias), with an **information-gain reward** (scouting); (4) returning the mixed move, the expected win-% *delta* of each option (calibrated EV), and a plain-English reason. It predicts what's *actually* predictable — the opponent's move, whether a line KOs, the value of a decision — **never** the match winner from preview.

6 · How we stay honest this time

Every model ships a **proper-score metric (log-loss/Brier) with a confidence interval and an honest baseline**, persisted to JSON and gated in CI. No capstone until its inputs clear their bars. CHOMP's claim ('our brings beat human brings') is a **winnable** test we will actually run. And we present results as **expected value with calibrated confidence** — the sports-analytics standard — because in this format that is the honest, useful truth.

Foundations in the repo: docs/LITERATURE-v2.md (full study, with sources) · docs/ALAKAZAM-v2-spec.md (full architecture spec, dependency graph, acceptance bars).

Key sources: VGC-Bench (arXiv 2506.10326) · Metamon (arXiv 2504.04395) · DeepNash/R-NaD (arXiv 2206.15378) · CICERO/piKL (arXiv 2210.05492) · AlphaStar (DeepMind) · ReBeL/DeepStack/Libratus · sports xG/EPV.